A NEW FORM OF RE-ARRANGEMENT OF THE RAYLEIGH-SCHRÖDINGER PERTURBATION SERIES

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An improvement of convergence of the Rayleigh-Schrödinger perturbation series may be achieved by a "selfconsistent" transfer of all the excessively large (usually: diagonal) perturbations into denominators. We propose a new and much more universal realization of such a scheme for fully general, off-diagonal matrix forms of the zero-order Hamiltonians H_0. A formal core of our recipe lies in a de-diagonalization of Schrödinger operators $H_0 - E_0 I = (I + \Omega)F(I + \Omega^+)$ via triangular annihilation- and creation-like matrices Ω and Ω^+. Via a pre-partitioning and on- as well as off-diagonal re-arrangement of H_0's, the standard diagonal "unperturbed spectrum" $E_0 + F$ is "smeared" selfconsistently. With tridiagonal or band-matrix H_0's, our new prescription contains the recently proposed perturbation expansions with continued fractions as a special case.

1. Introduction

Hamiltonians H of complicated molecular or nuclear systems are currently being defined by their matrix elements in some suitable basis $\{|n\rangle\}_{n=0}^{\infty}$. One of the most natural methods of construction of their low lying states is then provided by perturbation theory [1]: Starting from any diagonal-matrix approximation

$$H \approx H_0 = \sum_{n=0}^{\infty} |n\rangle\, \varepsilon_n \langle n| \tag{1}$$

we may formally introduce a small parameter λ, expand

$$H = H(\lambda) = H_0 + \lambda H_1 + \lambda^2 H_2 + \dots \tag{2}$$

and solve the underlying Schrödinger bound-state problem

$$H\,|\psi\rangle = E\,|\psi\rangle \tag{3}$$

via power series

$$E = E_0 + \lambda E_1 + \lambda^2 E_2 + \dots \tag{4}$$

$$|\psi\rangle = |\psi_0\rangle + \lambda|\psi_1\rangle + \lambda^2|\psi_2\rangle + \dots \,. \tag{5}$$

With an additional "mean-field" approximation

$$\varepsilon_n \approx \langle n|\; H\; |n\rangle \tag{6}$$

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the whole — so called Rayleigh-Schrödinger — scheme proves very efficient for most diagonally dominated H's in practice.

Sometimes, a few exceptional off-diagonal matrix elements of λH_1 (or of $\lambda^2 H_2$ etc) still remain large. Then, a cure is easy — we may move the large terms from λH_1 in H_0 (or from λH_2 in H_1 etc) and perform a (numerical) pre-diagonalization of the new operator H_0. In this way, we return back to the diagonally dominated situation again.

Difficulties arise when the number of large off-diagonalities grows too much. Then, the general sparse-matrix approximants H_0 have to be treated in a strong-coupling regime (cf. the few-body anharmonic-oscillator example [2]) or via singular perturbation techniques (e.g., for forces with a repulsive core in nuclear physics [3]). In the present paper, we intend to describe a new perturbation scheme tailored just to similar quasi-nonperturbative situations.

2. General formalism

In place of eq. (1), the general sparse-matrix zero-order Hamiltonian or rather Schrödinger operator $H_0 - E_0 I$ will be considered here in the factorized form

$$H_0 - E_0 I = (I + \Omega) \begin{pmatrix} F_0 & 0 & \dots & & \\ 0 & F_1 & 0 & \dots & \\ 0 & 0 & F_2 & 0 & \dots \\ & & & \dots & \end{pmatrix} (I + \Omega^+) \tag{7}$$

where the cross $^+$ denotes the ordinary Hermitean conjugation. This equation generalizes the standard Rayleigh-Schrödinger diagonality postulate (1) and reproduces it with trivial $\Omega = 0$.

For the sake of simplicity, we restrict our attention to off-diagonal and upper triangular matrices Ω,

$$\Omega_{nn} = \Omega_{nn-1} = \dots = \Omega_{n0} = 0. \tag{8}$$

The related class of Hamiltonians H_0 proves very rich: Even with the single nonzero diagonal in Ω's we already get tridiagonal operators $H_0 - E_0 I$ and non-perturbative chain models. These models appear in various physical applications (*e.g.*, [4]) as well as in the Lanczos-inspired computations [5].

In the literature, perturbation formalism with a pre-factorization has already been proposed for tridiagonally dominated Hamiltonians H in ref. [6]. In its present generalization, our purpose is two-fold. Firstly, we intend to extend the range of the method from tridiagonal to virtually arbitrary non-diagonal H_0's (Sect. 2). Simultaneously, we shall also succeed in a significant simplification of technicalities (Sect. 3). We emphasize that in contrast to the opinion expressed in ref. [6], our new factorizative version of perturbation theory with non-diagonal H_0's does not necessitate the evaluation of any infinite auxiliary continued fraction(s) at all.

2.1 The solvability of factorized zero-order problems

In accord with ref. [5], factorizations (7) need not exist for all H_0's and/or E_0's. We shall demand

$$F_1 \neq 0,\ F_2 \neq 0,\ \dots\ , \tag{9}$$

therefore. In principle, such a restriction is purely technical and might be removed in different ways. Here, we shall use it just for the sake of simplicity.

In connection with the zero-order form of eq. (3),

$$H_0\ |\psi_0\rangle = E_0\ |\psi_0\rangle \tag{10}$$

eqs. (7) and (9) reduce the pertaining secular equation $\det(H_0 - E_0 I) = 0$ to an almost trivial condition

$$F_0 = 0. \tag{11}$$

This is our first key observation. Again, we shall see below (*cf., e.g.*, eq. (30)) that the role of eq. (11) will remain very formal and also amenable to a weakening if needed [7].

Equation (11) is important as a guarantee of the existence of nontrivial wave functions $|\psi_0\rangle$ [5]. In the present perturbative context, its validity becomes trivial once we drop the first line and column from eq. (7).

The subsequent explicit construction of the zero-order wave functions becomes easy, starting by a pre-multiplication of equation (10) by the matrix

$$(I + \Omega)^{-1} = I - \Omega + \Omega^2 - \Omega^3 + \dots \tag{12}$$

and by the inverse elements F_j^{-1}, $j = 1, 2, \dots$. Of course, the latter quantities remain finite due to our regularity assumption eq. (9).

As a result, the new matrix zero-order Schrödinger equation acquires an infinite-dimensional Hessenberg [5] form

$$Q(I + \Omega^+)|\psi_0\rangle = 0 \tag{13}$$

where

$$Q = I - |0\rangle\langle 0| = \sum_{n=1}^{\infty} |n\rangle\langle n|$$

is a projector. It may easily be checked to possess the following compact solution

$$\langle m|\psi_0\rangle = (-1)^m\ \langle 0|\psi_0\rangle \times \det P^{[m]}. \tag{14}$$

Here, for all $m = 1,\ 2,\ \dots$, the determinants and matrices

$$P^{[m]} = \begin{pmatrix} (\Omega^+)_{1,0} & 1 & 0 & \dots & 0 \\ (\Omega^+)_{2,0} & (\Omega^+)_{2,1} & 1 & 0 & \dots \\ & & \dots & & \\ (\Omega^+)_{m-1,0} & \dots & (\Omega^+)_{m-1,m-3} & (\Omega^+)_{m-1,m-2} & 1 \\ (\Omega^+)_{m,0} & \dots & (\Omega^+)_{m,m-3} & (\Omega^+)_{m,m-2} & (\Omega^+)_{m,m-1} \end{pmatrix}$$

are finite-dimensional.

We may conclude that due to the simple postulate (11) and the — much more complicated — factorization assumption (7), our zero-order equation (10) is solvable non-numerically.

2.2 The formulae for the higher-order corrections

The pair of universal Rayleigh-Schrödinger power-series Ansaetze (4) and (5) converts our given Schrödinger equation (3) into its zero-order counterpart (10) accompanied by its subsequent $O(\lambda^k)$ systematic improvements

$$(H_0 - E_0 I)\ |\psi_k\rangle + (H_k - E_k)\ |\psi_0\rangle + |\tau_{k-1}\rangle = 0, \quad k = 1,\ 2,\ \ldots . \tag{15}$$

Here,

$$|\tau_{k-1}\rangle = (H_{k-1} - E_{k-1})|\psi_1\rangle + (H_{k-2} - E_{k-2})|\psi_2\rangle + \ldots (H_1 - E_1)|\psi_{k-1}\rangle \tag{16}$$

at $k > 1$ and $|\tau_0\rangle = 0$ at $k = 1$. Often, we omit $H_2 = H_3 = \ldots = 0$.

2.2.1 Energies

In full analogy with section 2.1, we may divide each subsequent $k > 0$ eq. (15) by the regular matrix $(I + \Omega)$ from the left. We get

$$\begin{pmatrix} 0 & 0 & \ldots & \\ 0 & F_1 & 0 & \ldots \\ & & \ldots & \end{pmatrix} (I+\Omega^+)|\psi_k\rangle + (I+\Omega)^{-1}[(H_k - E_k)|\psi_0\rangle + |\tau_{k-1}\rangle] = 0. \tag{17}$$

Obviously, the zeroth row of this equation does not contain any component of the k-th wave function correction $|\psi_k\rangle$ and becomes an explicit definition of the new energy,

$$E_k = \frac{\langle 0|(I+\Omega)^{-1} H_k |\psi_0\rangle + \langle 0|(I+\Omega)^{-1}|\tau_{k-1}\rangle}{\langle 0|(I+\Omega)^{-1}|\psi_0\rangle}. \tag{18}$$

Expansion (12) proves very useful here — our auxiliary matrix factors Ω have a property

$$(\Omega^m)_{n,n+m-1} = (\Omega^m)_{n,n+m-2} = \ldots = (\Omega^m)_{n,0} = 0, \quad m = 1,\ 2,\ \ldots \tag{19}$$

i.e., they become nilpotent after any finite truncation.

In full analogy with the standard textbook constructions [1], we may now treat E_k as an abbreviation, and solely pay attention to the recurrent evaluation of corrections to the wave functions.

2.2.2 Wave functions

Let us remind the reader that equations (15) imply the well known ambiguity

$$|\psi_k\rangle \to |\psi_k\rangle + \text{const}\ |\psi_0\rangle \tag{20}$$

of the wave functions. This entitles us to accept the normalization $|\psi_k\rangle = Q|\psi_k\rangle$, *i.e.*,

$$\langle 0|\psi_k\rangle = 0, \quad k = 1, 2, \ldots, \tag{21}$$

and, having already used the first row of eq. (17), re-write its rest in the new matrix form

$$\begin{pmatrix} F_1 & 0 & \dots & \\ 0 & F_2 & 0 & \dots \\ & & \dots & \end{pmatrix} (Q + Q\Omega^+ Q) \begin{pmatrix} \langle 1|\psi_k\rangle \\ \langle 2|\psi_k\rangle \\ \dots \end{pmatrix} = \begin{pmatrix} \langle 1|\rho\rangle \\ \langle 2|\rho\rangle \\ \dots \end{pmatrix} \tag{22}$$

with

$$|\rho\rangle = -(I+\Omega)^{-1}[(H_k - E_k)|\psi_0\rangle + |\tau_{k-1}\rangle]$$

and abbreviation (18) for E_k.

The latter equation contains non-zero matrix indices only. As a recurrent definition of $|\psi_k\rangle = Q|\psi_k\rangle$ it may be given the final and compact form

$$\langle m+1|\psi_k\rangle = \langle m+1|[Q - Q\Omega^+Q + (Q\Omega^+Q)^2 - \dots + (\pm 1)^m (Q\Omega^+Q)^m] \times \\ \times \begin{pmatrix} 1/F_1 & 0 & \dots & \\ 0 & 1/F_2 & 0 & \dots \\ & & \dots & \end{pmatrix} Q|\rho\rangle, \quad m = 0,\ 1,\ \dots \tag{23}$$

which, especially in a vicinity of diagonal H_0's, preserves the numerically important small-factor role of the unsmeared-propagator quantities F^{-1}.

2.3 An example of the smearings Ω

A key to applicability of the above perturbative recipe lies in the feasibility of factorization (7) of the zero-order Hamiltonian in question. For the sake of definiteness, let us restrict our attention to its simplest nontrivial (tridiagonal) example

$$H_0 = \begin{pmatrix} h_0 & a & 0 & \dots & & \\ a^\star & h_1 & b & 0 & \dots & \\ 0 & b^\star & h_2 & c & 0 & \dots \\ & & \dots & & & \end{pmatrix}. \tag{24}$$

Obviously, our factorization may now be introduced "by brute force", as an explicitly postulated identity

$$H_0 - E_0 I = \begin{pmatrix} 1 & u & 0 & \dots & & \\ 0 & 1 & v & 0 & \dots & \\ 0 & 0 & 1 & w & 0 & \dots \\ & & & \dots & & \end{pmatrix} \times \\ \times \begin{pmatrix} f_0 & 0 & \dots & & & \\ 0 & f_1 & 0 & \dots & & \\ 0 & 0 & f_2 & 0 & \dots & \\ & & & \dots & & \end{pmatrix} \times \begin{pmatrix} 1 & 0 & \dots & & & \\ u^\star & 1 & 0 & \dots & & \\ 0 & v^\star & 1 & 0 & \dots & \\ & & & \dots & & \end{pmatrix} \tag{25}$$

with some suitable parameter E_0.

Similar proposals were already made in the literature (cf. [6–8]). With abbreviations $u = a/f_1$, $v = b/f_2, \ldots$ this leads to the representation of matrix elements of the corresponding nontrivial, two-diagonal "smearings" in terms of a sequence of auxiliary continued fractions,

$$f_0 = h_0 - E_0 - \cfrac{a\, a^\star}{h_1 - E_0 - \cfrac{b\, b^\star}{h_2 - E_0 - \cfrac{c\, c^\star}{h_3 - \ldots}}}$$

$$f_1 = h_1 - E_0 - \cfrac{b\, b^\star}{h_2 - E_0 - \cfrac{c\, c^\star}{h_3 - \ldots}} \tag{26}$$

$$\ldots .$$

Of course, an evaluation of their values remains a purely numerical problem. Moreover, for the more general H_0's and Ω's, the use of similar recurrent brute-force methods must be preceded by an appropriate generalization of continued fractions [7].

3. The self-consistent construction of H_0's

Our present general factorization idea does not seem too appealing on the first sight. Indeed, in its preceding applications to some "realistic" Hamiltonians H with a strongly off-diagonal matrix structure (*cf.*, *e.g.*, [6–8] for illustrative examples), its feasibility required a (say, band-matrix or at least sparse-matrix) simplification of the underlying Schrödinger operators $H_0 - E_0 I$.

In the forthcoming climax of our paper, let us now describe one of the most natural "internally selfconsistent" realizations of our scheme in the entirely general case. A core of its idea will lie in the reversal of interpretation. Basically, we are going to construct the auxiliary Ω's first.

Our idea is rather new. Indeed, in all our older attempts which used non-diagonal H_0's, the preliminary transformation $H_0 \to \Omega$ was performed numerically or, in some simplest cases, in terms of certain auxiliary continued fractions. Thus, the (highly ambiguous) a priori choice of H_0's was followed by an (exact) computation of the (E_0-dependent) "smearings" Ω. Here, a direct introduction of these smearings will be strongly recommended.

In a more standard language, our new prescription is based on the well known $O(\lambda)$-ambiguity of perturbation theory. Thus, as long as each Hamiltonian H may be reduced to many alternative leading-order simplifications (such that $H_0[1] - H_0[2] = O(\lambda)$), all the sufficiently small $O(\lambda)$ re-arrangements of the split eq. (2) of H are always permissible. Vice versa, all the separate alternative candidates for H_0 may be considered equivalent to a "reference" operator $R \approx H$ such that $R - H_0[1] = O(\lambda)$, $R - H_0[2] = O(\lambda)$ etc. Now we are going to show that this overall $O(\lambda)$ ambiguity and re-arrangements may prove particularly useful for more complicated H_0's.

3.1 The reference zero-order Hamiltonians R

Whenever the reference matrices R remain non-diagonal for a finite number of basis states only, $H - R \approx$ a diagonal matrix, our factorization prescription will concern just finite-dimensional submatrices $R - E_0 I$, *i.e.*,

$$\begin{pmatrix} [R_{n,n} - E_0] & R_{n,n+1} & \dots & R_{n,n+m} \\ R_{n+1,n} & [R_{n+1,n+1} - E_0] & \dots & R_{n+1,n+m} \\ \dots & & \dots & \\ R_{n+m-1,n} & R_{n+m-1,n+1} & \dots & R_{n+m-1,n+m} \\ R_{n+m,n} & R_{n+m,n+1} & \dots & [R_{n+m,n+m} - E_0] \end{pmatrix}$$

of the full zero-order Schrödinger operators $H_0 - E_0 I$. In a "vectorial" notation, these matrices may be factorized easily,

$$\begin{pmatrix} g_n & a_1 & a_2 & \dots & & a_m \\ a_1^\star & g_{n+1} & b_1 & b_2 & \dots & b_{m-1} \\ \dots & & & & & \dots \\ a_{m-1}^\star & b_{m-2}^\star & \dots & & & z_1 \\ a_m^\star & b_{m-1}^\star & \dots & & & g_{n+m} \end{pmatrix} = \begin{pmatrix} 1 & u_1 & u_2 & \dots & & u_m \\ 0 & 1 & v_1 & v_2 & \dots & v_{m-1} \\ & & \dots & & & \\ 0 & \dots & & & 0 & 1 \end{pmatrix} \times$$

$$\times \begin{pmatrix} f_n^{[n,m]} & 0 & \dots & & 0 \\ 0 & f_{n+1}^{[n,m]} & 0 & \dots & 0 \\ & & \dots & & \\ 0 & \dots & & 0 & f_{n+m}^{[n,m]} \end{pmatrix} \times \begin{pmatrix} 1 & 0 & & \dots & 0 \\ u_1^\star & 1 & 0 & \dots & 0 \\ u_2^\star & v_1^\star & 1 & 0 & \dots \\ & & \dots & & \\ u_m^\star & v_{m-1}^\star & \dots & & 1 \end{pmatrix} \tag{27}$$

(*cf.* [7]).

In the next step, let us contemplate operators H and/or H_0 which are represented by the entirely general matrices even within our pre-determined and nonzero $O(\lambda)$ approximative precision. Then, using an arbitrary choice of the integers n and $m = m(n)$, we may still apply our above formula (27). Of course, we now have $R \approx H$ only for a submatrix. Nevertheless, within the given partition of a complete basis, quantities $f_k^{[n,m]}$ still carry a "compressed" physical information as contained in H.

In the light of refs. [6–8], the "compression of information" may be done — numerically — even in an infinite-dimensional limit and/or for purely diagonal H_0's. Thus, auxiliary quantities of various formalisms (*e.g.*, the unperturbed spectrum in the textbook perturbation recipes, or generalized continued fractions in our older constructions [6, 7]) re-appear now in our new language as mere special cases and/or limits.

In the numerical *practice*, with a limited (and often also pre-determined) precision, the actual infinite-dimensional limiting transitions (*cf.* [6]) are rarely needed of course. In this context, the forthcoming final paragraph of our paper is just going to emphasize that within the framework of perturbation theory, they are even redundant *in principle*.

3.2 Factorizations via perturbative re-arrangements and multiple partitioning of the Hamiltonians

The rigorous incorporation of errors via their consequent treatment as additional, "technical" perturbations represents just the key idea of our present paper. It remains easily feasible within the framework of the underlying perturbative philosophy itself.

Let us consider the growth of m's in eq. (27) first. Even when we cannot prove the convergence of quantities

$$U_n = u_k^{[0,m]}, \quad V_k = u_k^{[1,m]}, \quad W_k = u_k^{[2,m]}, \quad k = 1,\ 2,\ \ldots \tag{28}$$

and

$$F_k = f_k^{[k,m]}, \quad k = 1,\ 2,\ \ldots \tag{29}$$

to their vectorial continued fraction limits [7], we may still recommend the direct use of (of course, sufficiently large) finite values of integers m in our formulae.

One of the possible recipes reads as follows. In the first step, we pick up a suitable infinite-dimensional reference operator $R \approx H$. Such a 'preliminary' zero-order operator may differ from our final H_0 of course.

In the second step, we specify a set of suitable partitioning dimensions n and $m = m(n)$. For each of them, we factorize all our finite submatrices and make a suitable selection of "compressed" information (cf. eqs. (28) and (29)) from the resulting and, in general, highly redundant set of u's, v's and f's.

In the third step, we interpret these compressions as a new definition of some new, "optimal" or "maximally selfconsistent" factors Ω and F. Only from these latter factors, we finally re-construct our zero-order operator $H_0 - E_0 I$ and re-arranged perturbations ($\equiv H - H_0$).

In practice, all three steps may strongly depend on our intuition. Nevertheless, for the sake of definiteness, let us now make the following explicit recommendation and postulate our re-arranged zero-order propagators in the form

$$\begin{pmatrix} (H_0)_{00} - E_0 & (H_0)_{01} & (H_0)_{02} & \ldots & & \\ (H_0)_{10} & (H_0)_{11} - E_0 & (H_0)_{12} & \ldots & & \\ \ldots & & & & \ldots & \\ (H_0)_{k0} & (H_0)_{k1} & \ldots & (H_0)_{kk} - E_0 & (H_0)_{kk+1} & \ldots \\ \ldots & & & & \ldots & \end{pmatrix}$$

$$= \begin{pmatrix} 1 & U_1 & U_2 & \ldots & & U_m & \ldots \\ 0 & 1 & V_1 & V_2 & \ldots & V_{m-1} & \ldots \\ & & \ldots & & & & \end{pmatrix} \times$$

$$\times \begin{pmatrix} 0 & 0 & \ldots & & & \\ 0 & F_1 & 0 & \ldots & & \\ 0 & 0 & F_2 & 0 & \ldots & \\ 0 & 0 & 0 & F_2 & 0 & \ldots \\ & & & & \ldots & \end{pmatrix} \times \begin{pmatrix} 1 & 0 & \ldots & & & \\ U_1^\star & 1 & 0 & \ldots & & \\ U_2^\star & V_1^\star & 1 & 0 & \ldots & \\ U_3^\star & V_2^\star & W_1^\star & 1 & 0 & \ldots \\ & \ldots & & & & \end{pmatrix}. \tag{30}$$

This is precisely our required input factorization prescription (7). As long as it re-constructs our zero-order operator H_0 from its reference tentative predecessor R, it really is a re-arrangement prescription.

We may conclude: Via a slight modification of H_0's, the zero-order Schödinger operators become easily factorized in the way which, by construction, is extremely flexible. Its basic feature is that it leaves the "new" perturbations $\lambda H_1 \equiv H - H_0$ properly small: Their magnitude does not exceed the "old" order of magnitude $O(\lambda)$.

4. Summary

Most perturbation algorithms require exact knowledge of certain auxiliary quantities: Unperturbed spectra for $\Omega = 0$, continued fractions for one-diagonal Ω's [6], etc. In contrast to this, the so called selfconsistent prescriptions for perturbative solution of a given Schrödinger equation (3) proceed in an adaptive way. Here, we proposed a new one:

1. One of the main immanent limitations of selfconsistent considerations lies in their technical feasibility. Usually, we are only able to vary a small (most often, diagonal) part of the zero-order Hamiltonian H_0. A removal of this restriction is one of the main merits of our present methodical proposal.

2. Closed formulae of our $m < \infty$ approximants resolve some paradoxes encountered in our older constructions with non-diagonal H_0's [6–8]. Taking the tridiagonally dominated eq. (24) as an explicit example, its old perturbative (*i.e.*, inherently approximative) treatment needed all the quantities $U_1 = a/F_1$, $V_1 = b/F_2$, ... (continued fractions) defined as certain numerical limits. In the present new setting, finite approximants

$$f_n^{[n,m]} = h_n - E_r - \cfrac{a_1\ a_1^\star}{h_{n+1} - E_r - \cfrac{b_1\ b_1^\star}{h_{n+2} - \dots \ddots \dots - \cfrac{z_1\ z_1^\star}{h_{n+m} - E_r}}} \equiv F_n \tag{31}$$

with $n = (0), 1, 2, \dots$ suffice. They replace the old and infinite continued fraction expansions (26) which were unnecessarily difficult to compute.

3. In our old approaches to the strongly nondiagonal Hamiltonians, the occurrence of the general sparse matrices $R - E_0 I$ implied an even more pronounced numerical character of our perturbative results [8]. In the present language, a lot of flexibility is recovered. All the variations and changes in the auxiliary F's and/or Ω's remain simply tractable as additional perturbations.

4. Any physically motivated re-split (2) of H may be reflected by an $O(\lambda)$ modification of the intermediate reference operators R as well as of our re-constructed approximants H_0 (30) and perturbations $\lambda H_1 \equiv H - H_0$. Vice versa, in analogy with the mean-field theories (and, of course, in the same Hamiltonian-dependent way),

our mathematically motivated re-arrangements (which render our basic technicality — factorization — possible) might also find their "self-consistent" re-interpretation in the language of physics.

5. In practice, our choice of re-partitionings and m's will be closely related to the rate of the $m \to \infty$ convergence of our auxiliary capital-letter quantities (28) and (29) to their (= generalized continued fraction) limits. Thus, the estimates of precision and convergence for $m < \infty$ and $m = \infty$ will be very similar (*cf.*, *e.g.*, [7]).

6. The most important question of the overall convergence of perturbation series remains extremely difficult [8]. Indeed, equation (11) only defines the zero-order difference

$$\langle 0|H_0|0\rangle - E_0 = \Omega_{0,1}F_1(\Omega^+)_{1,0} + \Omega_{0,2}F_2(\Omega^+)_{2,0} + \dots \tag{32}$$

as a function of the variable E_0, and similar parameter-dependence will also concern all the other elements of H_0. Obviously, the rigorous proofs of convergence may hardly be expected in non-trivial cases.

7. A priori, our main idea of using the perturbative $O(\lambda)$ ambiguities in H_0 for solution of technical questions is highly appealing. Of course, detailed study of this new form of perturbative selfconsistencies represents an extensive program for future research. For the sake of convincibility, Table 1 is added here just as a very small quantitative illustration of such a program. In the Table, we may notice that

Table 1. The higher-order weakening of sensitivity of energies to their input estimate E_0 for anharmonic oscillator [6] ($E^{(\mathrm{exact})} \approx 4.649$, $N = -\log|E^{(\mathrm{pert.})} - E^{(\mathrm{exact})}|$).

The order of perturbation	The number of correct digits N			
E_0	3.00	2.00	0.00	−2.0
0	−2.0	−0.4	−0.7	−0.8
1	1.47	1.15	0.80	0.61
2	2.18	1.80	1.31	1.02
5	2.81	2.80	2.60	2.22
9	3.05	3.05	3.06	3.07

the "permissible interval" of selfconsistent changes (here, of the single parameter E_0) proved surprisingly large for the chosen example. It also quickly grows with the increasing perturbation order.

8. Our example does not seem exceptional — *cf.* also a few other similar tests in ref.[8]. In their light, we may generalize their conclusions and expect that our basic "feasibility modifications" $H \to R \to H_0$ will not significantly worsen the precision obtainable, in principle (*e.g.*, numerically), from R itself.

9. The main improvements of precision may be expected to follow from the higher flexibility and improvements of R's themselves. Unfortunately, the thorough tests

of this expectation lie far beyond the scope of the present paper and have to be performed in the future. The corresponding motivation and models might be sought in molecular physics and quantum chemistry where various non-diagonal approximate Hamiltonians often occur in the standard Hückel models [9] and their various generalizations [10]. Preliminary considerations in this direction are encouraging [11].

10. Mathematically, our multiple truncation scheme exhibits certain "parallel-computing" features which could prove desirable in an explicit numerical algorithm. Thus, our prescription might also find some (presumably, purely computational) applications at the small and even very small partitions (*i.e.*, small m's).

11. In the non-computational and more conceptual considerations, the elementary partitioning and re-construction of H_0 exhibit a close relationship to the "mean-field" techniques and/or estimates (*cf.*, *e.g.*, (6)) and their generalizations. Hence, in principle, our formalism might also inspire some new independent insights in the underlying physics itself.

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